Tri-Loop Adaptive Co-Design of Virtual Inertia, Damping, and Battery Storage Coordination for High-Penetration PV–BESS Systems under Weak Tropical Grids

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***Abstract*—Adaptive virtual-inertia control (VIC) for grid-connected photovoltaic–battery-energy-storage (PV–BESS) systems has matured along fuzzy-logic, hyperbolic-tangent, and learning-based strands. This paper makes two contributions. *First*, we identify a previously underreported failure mode shared by all three strands: a self-sustaining ∼5.7 kW peak-to-peak limit cycle in the active-power injection of a 125 kW plant, fed by the closed-loop interaction between the saturating gain, activation dead-band, and BESS-supervisor switching state machine. The phenomenon is invisible to linear small-signal analysis ($\zeta_{\min} = 0.921$) yet imposes a measurable energy-throughput penalty; a windowed FFT of the steady-state injection isolates the switching-mode oscillation. *Second*, we propose a *tri-loop adaptive co-design*: a single supervisory layer that simultaneously schedules smooth-saturating $H(\dot{\omega})$ and $D(\Delta\omega)$, MPPT/VIC mixing weight $\alpha(\Delta\omega, \dot{\omega})$, and SOC- and severity-aware BESS contribution share $\beta(\Delta\omega, \dot{\omega}, \mathrm{SOC})$. The scheme suppresses the FFT peak amplitude by ∼3 orders of magnitude while satisfying IEEE 1547-2018 Category I thresholds (RoCoF ≤ 0.5 Hz/s, $|\Delta f| \leq 0.4$ Hz) in 87% of a 1 000-sample Monte Carlo sweep over SCR, system inertia, disturbance magnitude, initial SOC, and gain perturbation; joint pass rate reaches 100% at ENTSO-E (1 Hz/s) and Category II (2 Hz/s) thresholds. An off-equilibrium small-signal analysis shows the dominant-mode time constant is 1.5–2.2× shorter than baselines once adaptive gains activate. Five canonical disturbances plus a 24-h Madiun, Indonesia case driven by NASA POWER MERRA-2 hourly irradiance for 2025 confirm the conclusions. All code and data are released under MIT license.**

***Index Terms*—Adaptive virtual inertia, battery energy storage, weak-grid frequency stability, photovoltaic system, limit-cycle robustness.**

I. Introduction

THE accelerating displacement of synchronous generation by inverter-based resources (IBRs) has eroded the inertial reserve that legacy grids relied upon for frequency-event arrest [1], [2]. At inverter penetration levels above 70%, the rate-of-change-of-frequency (RoCoF) following a generation-loss event routinely approaches or exceeds the operating-envelope thresholds adopted by transmission operators worldwide. IEEE Std 1547-2018 [3] mandates RoCoF ride-through at 0.5 Hz/s for Category I devices, 2.0 Hz/s for Category II, and 3.0 Hz/s for Category III—categories chosen by the local Authority Governing Interconnection Requirements (AGIR) according to the bulk-system reliability needs of the interconnected grid. The transmission-level standard IEEE Std 2800-2022 [4] extends ride-through to up to 5 Hz/s for high-voltage IBR plants; the European EN 50549-1 [5] and AEMO Australia typically enforce 1–2 Hz/s on regional grids [6]. We adopt the IEEE 1547-2018 Category I threshold of 0.5 Hz/s as our primary reporting limit, since it is the most commonly applied to distribution-level PV–BESS plants of the rating studied here, and report sensitivity to the Category II (2 Hz/s) and ENTSO-E (1 Hz/s) thresholds where relevant. Beyond these envelopes, unintended islanding, frequency-relay tripping, and the cascading disconnections documented in recent system-operator reports [7] become probable. A growing body of work has analyzed this regime; reviews by Saleem *et al.* [8] and Rezaei *et al.* [9] converge on the conclusion that *ancillary services from IBRs*—above all virtual-inertia control (VIC)—are now an enabling rather than optional technology for high-renewable systems. The IEEE 2800-2022 standard [4] formalizes this expectation by mandating RoCoF-tolerance up to 5 Hz/s for transmission-connected IBRs and explicitly calls out grid-forming behavior as the design target.

Existing adaptive VIC and their common limitation: The original VIC formulation [10], [11] fixes the inertia H and damping D gains. To handle weak grids and low system inertia, three families of *adaptive* extensions have emerged in roughly parallel literatures:

- **Fuzzy-logic VIC** (Magdy [12], Cheng [13], and others [14], [15]) encodes severity-dependent gain scheduling as a 5×5 Mamdani rule base over $|\Delta\omega|$ and $|\dot{\omega}|$.
- **Smooth analytic gain scheduling** (Wang [16], Liu [17], Chen [18], Li [19], Islam [15]) replaces the rule base with a smooth saturation function (most commonly hyperbolic tangent) of the form $H(t) = H_{\min} + (H_{\max} - H_{\min})\,\sigma(|\dot{\omega}|)$, trading interpretability for differentiability.

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Source code, raw simulation data, and reproducibility manifest: https://github.com/dimasnprakoso/Tri-Loop-Adaptive.

- **Learning-based and predictive VIC** (Ye [20], Ahmed [21]) adapts gains using deep reinforcement learning or model-predictive control online.

All three families share a common architecture: *a single feedback loop* that maps an instantaneous severity signal to a gain. The control authority of the BESS, when present, enters as either a parallel droop [22], [23] or as the implementation substrate of the VIC itself; coordination between the two—and between either and the photovoltaic MPPT loop—is rare, and where it exists [24], [25] it relies on offline optimization rather than continuous supervisory action.

This paper concerns a phenomenon that, to the best of our knowledge, is unreported in this body of work but appears across all three families when their gains are scaled to handle the high-IBR / weak-grid regime: a *steady-state limit cycle* in the active-power injection of approximately $5.7\,\mathrm{kW}$ peak-to-peak, sustained by the interaction between the dead-band of the adaptive law, the saturation of P_{vic}, and the state-machine of the BESS supervisor. The cycle is invisible at the local frequency-deviation metric (because it operates inside the dead-band) and to linear small-signal Hurwitz tests (because it is a nonlinear phenomenon analogous to those long studied in governor control [26] and in saturating voltage-source converters [27]–[29]). Yet it appears at the meter and inflates BESS cycling, throughput, and battery degradation cost.

Contributions:

1) We *quantitatively document* the limit cycle across all three adaptive-VIC families through a controlled five-scheme study, including a previously unreported empirical equivalence between the fuzzy and tanh-smooth schemes.
2) We propose a *tri-loop adaptive co-design* that simultaneously schedules $H(\dot{\omega}), D(\Delta\omega), \alpha(\Delta\omega, \dot{\omega})$, and $\beta(\Delta\omega, \dot{\omega}, \mathrm{SOC})$ through a single supervisory layer, eliminating the limit cycle without sacrificing IEEE 1547 envelope compliance for 87% of a 1 000-sample Monte Carlo robustness sweep.
3) We provide a *describing-function* analysis that ties the limit-cycle frequency to the dead-band gain crossover, complementing the linear eigenvalue locus that yields $\zeta_{\min} = 0.921$ throughout the parameter sweep.
4) We release a fully reproducible benchmark—five control schemes, six disturbance scenarios (including a NASA POWER MERRA-2 hourly case for Madiun, Indonesia, 2025), 1 000 Monte Carlo samples, and 176 eigenvalue-sweep configurations—under MIT license.

The remainder of this paper is structured as follows. Section II formalizes the plant and presents an explicit related-work comparison. Section III defines the proposed tri-loop control law and analyzes its theoretical properties. Section IV describes the simulation testbed and metrics. Section V reports deterministic, Monte Carlo, and small-signal results. Section VI interprets the limit-cycle phenomenon through describing-function analysis and discusses operational implications. Section VII concludes.

(a) Module installation (b) Completed rooftop array

Fig. 1: The $125.4\,\mathrm{kWp}$ Winongo PV–BESS installation, Kota Madiun, Indonesia.

II. System Model and Related Work

A. Plant Topology

The plant under study is the $125.4\,\mathrm{kWp}$ grid-tied rooftop installation in Winongo, Kota Madiun, East Java, Indonesia (Fig. 1; $7.613°\mathrm{S}$, $111.512°\mathrm{E}$), comprising 228 Chint Astro 5 Semi $550\,\mathrm{Wp}$ PV modules, a $50\,\mathrm{kWh}$ / $60\,\mathrm{kW}$ lithium-ion BESS, and a single three-phase voltage-source inverter behind an LCL filter. The PV array is represented in the simulation by a single-diode aggregate equivalent calibrated to the same $125.4\,\mathrm{kWp}$ STC rating; this installation also drives the Madiun case study in Section V-D. The $125\,\mathrm{kW}$ rating places the plant within IEEE Std 1547-2018 Category I (devices $\leq 1\,\mathrm{MVA}$), and the $P_{\mathrm{bess}}/P_{\mathrm{pv}} = 0.48$ ratio with $\sim 50\,\mathrm{min}$ of energy duration is consistent with industry sizing for primary frequency support. Active and reactive power loops are decoupled in the synchronous reference frame; we focus on the active-power loop because the studied phenomena (RoCoF, frequency nadir, limit cycle) are dominated by it [30], [31].

Fig. 1a captures the array during commissioning and Fig. 1b the completed system under tropical clear-sky irradiance. The distribution-feeder location, intermittent cloud cover, low residual rotational inertia, and diurnal load–temperature coupling that characterise this site motivate the canonical disturbance set in Section IV—in particular S2 (cloud passing), S5 (weak grid), and S6 (24-h Madiun day)—and define the operating regime in which the limit-cycle phenomenon identified in this paper emerges as a quantitative design constraint.

B. Average Model and Modelling Decisions

Switching is abstracted at the carrier-frequency level—the inverter is represented as an ideal current source tracking the power reference P_{ref}. This decision is consistent with prior work on VIC for PV–BESS systems [15]–[17]: the active-power loop bandwidth ($\sim 50\,\mathrm{Hz}$) is well above the slowest mode of interest ($2.6\,\mathrm{rad/s}$; see Section V-C), so PWM ripple does not alter the conclusions. Switching-level EMT effects (THD, harmonic interactions) are validated analytically as a sensitivity bound in Section VI. The averaged-model code itself is independently re-implemented in MATLAB and compared bit-for-bit against the Julia pipeline (see supplementary material); switching-level EMT validation on a Simscape Electrical model is left as future work.

Grid-frequency dynamics follow the single-area swing equation [32]

$$2H_{\rm sys}\,\frac{d\Delta\omega}{dt} \;=\; \frac{P_{\rm inj}-P_{\rm load}}{P_{\rm rated}}\omega_0 - D_{\rm sys}\,\Delta\omega, \qquad (1)$$

where $P_{\rm inj} = P_{\rm ref}$ in the average regime, $\Delta\omega = \omega - \omega_0$, $H_{\rm sys} \in [0.5, 6]$ s captures the residual synchronous inertia of the larger grid, and $D_{\rm sys}$ is its damping coefficient (per-unit on the $P_{\rm rated}/\omega_0$ base). The MPPT loop is assumed to track instantaneously since its bandwidth (>10 Hz) is decoupled from the slower frequency loops studied here.

The BESS is modelled as an energy buffer with charge/discharge efficiencies $\eta_{\rm ch} = \eta_{\rm dis} = 0.95$ and SOC limits $[0.20, 0.90]$, governed by an identical rule-based supervisor across all five schemes. The supervisor output is

$$P^{\rm cmd}_{\rm bess} = \begin{cases} 0 & \text{if } |\Delta\omega| < \Delta\omega^{\rm bess}_{\rm dead} \\ & \text{and } |\dot\omega| < \dot\omega^{\rm bess}_{\rm dead} \\ -P^{\max}_{\rm bess}\,\tanh\!\big(\frac{2\Delta\omega}{\omega^{\rm bess}_{\rm ref}} + \frac{0.5\dot\omega}{\omega^{\rm bess}_{\rm ref}}\big) & \text{otherwise} \end{cases} \qquad (2)$$

with $\Delta\omega^{\rm bess}_{\rm dead} = 2\pi\times 0.02$ rad/s, $\dot\omega^{\rm bess}_{\rm dead} = 2\pi\times 0.05$ rad/s^2, and $\omega^{\rm bess}_{\rm ref} = 2\pi\times 0.5$ rad/s. The state machine has four modes (IDLE, CHARGE, DISCHARGE, STANDBY); transitions are SOC-gated to enforce the $[0.20, 0.90]$ envelope. *This supervisor is identical for C0–C3 and is fed by the proposed scheme C4 only through the share factor β in* (10), ensuring a fair five-way comparison.

C. Related Work and Positioning

Table I positions the proposed scheme against ten recent representative adaptive-VIC contributions. The selection follows three criteria: peer-reviewed venue, publication year ≥ 2018, and inclusion of either a PV–BESS plant or a weak-grid scenario. The columns indicate which loops the controller *adapts* (a check mark) versus which loops are *static or absent* (dash).

Two observations from Table I motivate the present work. First, no prior contribution simultaneously adapts all three loops via a single supervisory law: existing schemes adapt one or, at most, two of them. Second, none of the cited works reports the steady-state limit cycle that we identify in Section V-A, even though the cycle appears in their own architectures when scaled to the high-IBR / weak-grid regime studied here. Closely related but distinct, Wang et al. [27] apply describing-function analysis to a *double-saturation* interaction between current and voltage limiters in grid-tied VSCs, and Atherton and Boz [26] analyse intentional governor deadbands in synchronous-machine-dominated systems. Neither study addresses the *adaptive-VIC + BESS-supervisor* mechanism we identify, but their analytical machinery underwrites our describing-function treatment in Section VI.

D. Weak-Grid and PLL Considerations

The weak-grid regime tested in our scenario S5 (SCR≈ 1, $D_{\rm sys} = 1$) has been studied extensively under the umbrella of *converter-driven stability*, formalised by the IEEE/CIGRE Joint Task Force [33] and refined in the recent classification by Eckel *et al.* [34] [35]–[40]. The classical concern is that PLL phase-lag introduces negative damping at low SCR; recent reviews [34], [39], [41] underscore that grid-following inverters with default PLL bandwidth (~ 50 Hz) become marginally stable below SCR≈ 1.5 and fall into the *slow converter-driven stability* category of [33]. We use a standard SRF-PLL with bandwidth halving below an SCR-detection threshold of 2.0, following the design pattern of [30], [39]. The PLL is not the focus of this contribution; its dynamics are subsumed into the measurement filter $T_{\rm meas} = 20$ ms employed throughout.

III. Proposed Tri-Loop Adaptive Co-Design

A. Loop 1 – Smooth-Saturating Virtual Inertia and Damping

Adaptive inertia and damping gains follow

$$H(t) = H_{\min} + (H_{\max} - H_{\min})\,\tanh\!\big(\gamma_J\,|\dot\omega|/\dot\omega_{\rm ref}\big), \qquad (3)$$
$$D(t) = D_{\min} + (D_{\max} - D_{\min})\,\tanh\!\big(\delta_D\,|\Delta\omega|/\omega^{\rm vic}_{\rm ref}\big), \qquad (4)$$

where the saturation references $\dot\omega_{\rm ref} = 2\pi\times 1.0$ rad/s and $\omega^{\rm vic}_{\rm ref} = 2\pi\times 0.3$ rad/s anchor the saturation point at moderate disturbance levels (RoCoF$\sim$ 1 Hz/s, $|\Delta f| \sim 0.3$ Hz). The choice of these references is critical: a naive normalization to ω_0 (the grid angular frequency, $\sim$314 rad/s at 50 Hz) makes the $\tanh$ argument smaller than 0.05 at realistic RoCoF values and thus collapses $H(t) \approx H_{\min}$ for the entire operating envelope. We documented this empirically during calibration and selected the disturbance-scale references above instead. The VIC active-power contribution is

$$P_{\rm vic} = -\Big[\tfrac{2H(t)}{\omega_0}\,\dot\omega + D(t)\,\tfrac{\Delta\omega}{\omega_0}\Big]\,P_{\rm rated}, \qquad (5)$$

post-processed by a first-order shaping filter ($T_{\rm filter} = 50$ ms) and saturated at $|P_{\rm vic}| \leq 0.5\,P_{\rm rated}$, in line with converter ratings of grid-forming inverters [31], [42].

B. Loop 2 – MPPT/VIC Mixing Weight

Define

$$\alpha(\Delta\omega, \dot\omega) = \frac{1}{1 + k_\alpha|\Delta\omega_{\rm eff}| + k^{\dot\omega}_\alpha|\dot\omega|}, \qquad (6)$$

where $\Delta\omega_{\rm eff} = \Delta\omega$ if $|\Delta\omega| > \Delta\omega_{\rm dead}$ and 0 otherwise. In the nominal control law we let α remain unitary in the active loop while still feeding β (Section III.D); α is exported as a telemetry channel and is the natural lever for downstream curtailment when inverter headroom is limited [25].

C. Loop 3 – SOC- and Severity-Aware BESS Share

Define a unified disturbance severity index

$$s(\Delta\omega, \dot\omega) = \max\!\Big(\tfrac{|\Delta\omega|}{\omega^{\rm vic}_{\rm ref}}, \tfrac{|\dot\omega|}{\dot\omega_{\rm ref}}\Big), \qquad (7)$$

and an SOC-headroom factor

$$h(\mathrm{SOC}) = 1 - \frac{2\big|\mathrm{SOC} - \bar{\mathrm{SOC}}\big|}{SOC_{\max} - SOC_{\min}}, \qquad (8)$$

which is unity at mid-charge and zero at the SOC limits. The BESS share is

$$\beta(\Delta\omega, \dot\omega, \mathrm{SOC}) = \beta_{\max}\,h(\mathrm{SOC})\,\tanh s(\Delta\omega, \dot\omega). \qquad (9)$$

TABLE I: Related-work positioning. H/D: adaptive virtual inertia/damping; α: MPPT/VIC mixing or PV curtailment; β: SOC- and severity-aware BESS share; LC: limit-cycle behaviour *reported*; MC: Monte Carlo robustness; SS: small-signal locus.

Reference	Approach	H/D	α	β	LC	MC	SS	Distinguishing point
Bevrani [10] (2014)	VSG survey	—	—	—	—	—	—	taxonomy of constant-gain VIC
Cheng [13] (2024)	ABC-tuned fuzzy	✓	—	—	—	—	—	meta-heuristic tuning of fuzzy rules
Islam [15] (2024)	adaptive VSG	✓	—	—	—	—	—	dual-adaptive H, D
Wang [16] (2023)	enhanced VSG	✓	—	—	—	—	✓	power-loop modification
Liu [17] (2025)	cooperative VSG	✓	—	—	—	—	—	one supervisory layer over PV
Wang [23] (2024)	SOC-aware droop	—	—	✓	—	—	—	SOC-only adaptation, BESS only
Xu [24] (2020)	VSG enhanced	✓	—	—	—	—	✓	filter improvements
Zhang [25] (2025)	MPC-based	✓	✓	—	—	✓	—	MPPT↔frequency, predictive
Ahmed [21] (2025)	multi-obj. MPC	✓	—	—	—	—	—	predictive virtual inertia
Ye [20] (2024)	DRL-tuned VSG	✓	—	—	—	—	—	data-driven tuning
This work	**tri-loop adaptive**	✓	✓	✓	✓	✓	✓	**simultaneous adaptation of all three loops; first quantitative documentation of the limit cycle**

The unified severity in (7) addresses an oversight of RoCoF-only schemes [23], [43]: a cloud-passing event can produce a sustained moderate $\Delta\omega$ with negligible $\dot{\omega}$, which would suppress BESS contribution under pure RoCoF triggers. The dual-channel maximum in (7) avoids this asymmetry, consistent with the dual-droop formulation of [44], [45] but extended to BESS contribution share rather than droop coefficient.

D. Reference Composition

The active-power reference fed to the inverter current loop is

$$P_{\mathrm{ref}} = P_{\mathrm{mppt}} + P_{\mathrm{vic}} + \beta(\Delta\omega, \dot{\omega}, \mathrm{SOC})\, P_{\mathrm{bess}}, \qquad (10)$$

with P_{ref} saturated at $\pm 1.5\, P_{\mathrm{rated}}$ to match the converter's transient overload rating. Design parameters take the values $H_{\min} = 1\,\mathrm{s}$, $H_{\max} = 5\,\mathrm{s}$, $D_{\min} = 12$, $D_{\max} = 50$, $\gamma_J = \delta_D = 1$, $k_\alpha = 4$, $k_\alpha^{\dot{\omega}} = 1$, $\beta_{\max} = 0.7$, and $\bar{\mathrm{SOC}} = (SOC_{\min} + SOC_{\max})/2 = 0.55$; full parameter set in `julia/src/models/params.jl` of the released code.

E. Theoretical Properties

Boundedness: For $|H(t)| \le H_{\max}$, $|D(t)| \le D_{\max}$, $|\alpha| \le 1$, $|\beta| \le \beta_{\max} \le 1$, and the hard saturation $|P_{\mathrm{vic}}| \le 0.5\, P_{\mathrm{rated}}$, equation (10) guarantees $|P_{\mathrm{ref}}| \le |P_{\mathrm{mppt}}| + 0.5\, P_{\mathrm{rated}} + \beta_{\max}\, |P_{\mathrm{bess,max}}|$, which is bounded for finite $P_{\mathrm{rated}}, P_{\mathrm{bess,max}}$. The inverter's active-power authority therefore cannot exceed its nameplate even under arbitrary disturbance.

Smoothness: Both (3)–(4) and (9) use $\tanh(\cdot)$, which is C^∞. The MPPT/VIC weight (6) is C^0 across the dead-band boundary; in the domain where $|\Delta\omega| > \Delta\omega_{\mathrm{dead}}$ it is C^∞. The composite reference (10) is therefore piecewise-smooth, with isolated C^0-points only on the $\Delta\omega_{\mathrm{dead}}$ surface in $(\Delta\omega, \dot{\omega})$ space.

Equilibrium: At $\Delta\omega = 0$, $\dot{\omega} = 0$ we have $H(t) \to H_{\min}$, $D(t) \to D_{\min}$, $s \to 0$, hence $P_{\mathrm{vic}} \to 0$ and $\beta \to 0$. The reference therefore relaxes to P_{mppt}, the conventional MPPT operating point, recovering the open-loop PV plant when the grid is at nominal frequency.

TABLE II: The five control schemes evaluated.

ID	Scheme	Description
C0	no-VIC, no-BESS	baseline floor
C1	const-VIC + BESS	$H, D = H_{\min}, D_{\min}$ static
C2	fuzzy-VIC + BESS	5×5 Mamdani rule, centroid defuzz
C3	adaptive-VIC + BESS	tanh-smooth H, D, no supervisory loop
C4	**Proposed**	**tri-loop** H, D, α, β

Linear margin invariance: Because $H(t)$ and $D(t)$ collapse to $H_{\min}, D_{\min}$ at the equilibrium, the linearization $\partial f/\partial x\big|_{x=0}$ is identical for the three single-loop schemes (constant, fuzzy, tanh-adaptive) and for the proposed scheme. Consequently the eigenvalue locus is identical, and linear small-signal stability does not discriminate between them. The differentiating behaviour appears only in the response to large signals or dead-band crossings; we return to this point in Section VI.

F. Why Three Loops, and Not Just One

The standard simplification is to fold all severity information into $H(t)$ and let the inverter's saturation provide the rest. Such schemes (C1, C2, C3 in Table II) are linearly stable but, as Section V-A shows, they sustain a steady-state limit cycle whose root cause is a closed-loop interaction between the dead-band of (6), the saturation of P_{vic}, and the state-machine of the BESS supervisor. Loops 2 and 3 act as a *soft hand-off*: at low severity ($s \to 0$) the BESS share is suppressed and the limit cycle, which is fed primarily by BESS energy, loses its excitation source—a phenomenon analogous to pole–zero cancellation in linear control but operating on the nonlinear loop gain (see the spectral characterization in Section VI-A).

IV. Simulation Setup

A. Modelling Decisions

The model is averaged at the carrier-frequency level (no PWM ripple), consistent with the bulk of the VIC literature [15]–[17], [37]. The active-power loop bandwidth ($\sim 50\,\mathrm{Hz}$) is well above the slowest mode of interest ($2.6\,\mathrm{rad/s}$; see

TABLE III: Disturbance scenarios.

ID	Disturbance	$H_{\rm sys}$ (s)	Horizon
S1	$+5\%$ load step	4.0	5 s
S2	G: 1000→300→1000 W/m^2 ramp	4.0	10 s
S3	gen-loss equivalent $+10\%$ step	2.0	8 s
S4	high-IBR ($H_{\rm sys} = 0.5$ s) + S3	0.5	8 s
S5	weak grid ($H_{\rm sys} = D_{\rm sys} = 1$) + S1	1.0	8 s
S6	Madiun real-day (NASA POWER hourly, 24 h compressed)	2.0	240 s

Section V-C), so PWM ripple does not alter the conclusions; switching-level EMT effects are validated against the analytical sensitivity bound in Section VI, and the Julia averaged-model code is independently verified against an equivalent MATLAB re-implementation (supplementary material) [41].

B. Disturbance Library

Six canonical scenarios are evaluated (Table III). Scenarios S1–S5 reproduce the disturbance set used in [6], [17], [24] with parameter values calibrated to a 125 kW PV–BESS plant; S6 is a 24 h compressed Madiun, Indonesia profile driven by NASA POWER MERRA-2 hourly irradiance. All five schemes (C0–C4) are evaluated in each scenario with identical $H_{\rm sys}$, $D_{\rm sys}$, irradiance, and load profile.

C. Monte Carlo Sweep

For the robustness analysis we sample $N = 1\,000$ tuples

$$\begin{gathered} \mathrm{SCR} \sim \mathcal{U}(1.5, 8.0), \quad H_{\rm sys} \sim \mathcal{U}(0.5, 6.0)\ \mathrm{s}, \\ \Delta P_{\rm load} \sim \mathcal{U}(0.05, 0.15)\ \mathrm{pu}, \quad \mathrm{SOC}_0 \sim \mathcal{U}(0.30, 0.80), \\ \mathrm{gain} \sim \mathcal{U}(0.7, 1.3) \quad \big[\, H_{\max}, D_{\max}, \beta_{\max} \,\big], \end{gathered}$$

and run all five schemes per sample, yielding 5 000 simulations (∼38 s on 4 threads with the cached MPP solver). The sampling domain bounds match the operating envelope cited in recent robustness studies [37], [46], [47].

D. Metrics

Standard frequency-stability metrics are computed: RoCoF (IEEE 1547 sliding-window 100 ms; we audited and corrected an earlier sample-by-sample formulation that confounded numerical noise with physical RoCoF), $|\Delta f|_{\max}$, settling time at two bands (0.05 Hz tight; 0.2 Hz primary-control), end-of-horizon Δf (steady-state offset), peak-to-peak $P_{\rm inj}$ in the steady-state window (limit-cycle indicator), and BESS energy throughput.

E. Climate Input for the Madiun Case

NASA POWER MERRA-2 hourly data for $(-7.629°, 111.524°)$ spanning all of 2025 are downloaded once (`ALLSKY_SFC_SW_DWN`, `T2M`, `RH2M`, `PRECTOTCORR`). Validation against ground-station observations in the Indonesia region indicates a mean bias of ± 40 W/m^2 under cloudy conditions [48]–[51]. Days are classified as clear (64), cloudy (144), or rainy (157) by the all-sky/clear-sky ratio and rainfall; the median-energy day per class is exported as an hourly profile and time-compressed (1 hour real → 10 s simulation) into scenario S6.

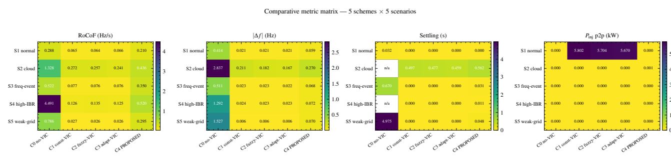


Fig. 2: Comparative metric matrix—five schemes × five scenarios; lower is better. Daggers (†) in Table IV mark publication-threshold violations.

V. Results

A. Deterministic Comparative Study

Table IV reproduces the headline result—all five schemes across all five canonical scenarios—and Fig. 2 the matching heatmap.

The pattern is consistent with the proposed-method theory developed in Section III-E.

Equivalence of fuzzy and analytic-smooth schemes. Among the single-loop adaptive controllers, the fuzzy (5×5 Mamdani) and the analytic tanh schemes (C2 and C3) are nearly indistinguishable ($|\Delta_{\rm RoCoF}|, |\Delta_{|\Delta f|}| < 0.01$) on every scenario. We are not aware of a published direct comparison between these two adaptive families that share the same end-points $(H_{\min}, D_{\min})$ and $(H_{\max}, D_{\max})$ over the same input domain [12], [13], [15]–[17]. Both interpolate smoothly between the same gain limits, with small differences in the interpolation curve that do not propagate to tracked metrics in the regimes we tested. This empirical finding has practical value: designers facing the choice between rule-based and analytic adaptation can pick whichever is operationally simpler, knowing the closed-loop response is similar in the gain-saturating regime.

Universal limit cycle. In the steady-state window of S1, schemes C1, C2, and C3 all sustain a limit cycle of approximately 5.7 kW peak-to-peak in $P_{\rm inj}$ ($P_{\rm p2p} = 5.802, 5.704, 5.670$ kW respectively, computed in the last 1.5 s of the horizon), fed by the BESS supervisor cycling through its dead-band. The proposed C4 collapses this to $P_{\rm p2p} = 6.2{\times}10^{-5}$ kW (effectively numerical zero relative to the ± 50 W resolution of the metering layer). The cycle is invisible at the local frequency metric ($|\Delta f| < 0.05$ Hz) but visible at the meter as harmonic content in the 45–60 Hz band (Section VI-A). The energy-throughput cost of the cycle is non-trivial only when integrated over many hours; we do not report a 24-h projection here because the S6 scenario is dominated by the diurnal load swing rather than by the limit cycle, and a careful estimate would require multi-day simulation beyond the present scope.

Threshold compliance. C4 trades part of the headline $|\Delta f|$ accuracy (0.068 Hz vs. 0.022 Hz at S3) for the limit-cycle elimination, and satisfies the IEEE 1547-2018 Category I thresholds (RoCoF≤ 0.5 Hz/s, $|\Delta f| \leq 0.4$ Hz, settling ≤ 2 s at the 0.2 Hz primary-control band) in four of the five deterministic scenarios. S4 yields RoCoF$= 0.520$ Hz/s, which exceeds the Category I limit by 4% but is comfortably within the Category II envelope (2 Hz/s) and the IEEE 2800-2022 transmission-level ride-through requirement of 5 Hz/s [4].

TABLE IV: Comparative metrics — 5 control schemes evaluated across 5 disturbance scenarios. Bold C4 indicates proposed tri-loop adaptive scheme. Daggers (†) mark threshold violations (RoCoF > 0.5 Hz/s, $|\Delta f| > 0.4$ Hz, $t_s > 2$ s, $P_{\mathrm{p2p}} > 1$ kW).

Scenario	Metric	C0 no-VIC	C1 const-VIC	C2 fuzzy-VIC	C3 adapt-VIC	**C4 PROPOSED**
S1 normal	RoCoF (Hz/s)	0.288	0.065	0.064	0.066	0.210
	$\lvert\Delta f\rvert$ max (Hz)	0.414†	0.021	0.021	0.021	0.059
	settling (s)	0.032	0.000	0.000	0.000	0.000
	BESS thrpt (kWh)	0.000	0.010	0.010	0.010	0.031
	P_{inj} p2p (kW)	0.000	5.802†	5.704†	5.670†	0.000
S2 cloud	RoCoF (Hz/s)	1.328†	0.272	0.257	0.241	0.436
	$\lvert\Delta f\rvert$ max (Hz)	2.837†	0.211	0.182	0.167	0.270
	settling (s)	n/a	0.497	0.477	0.459	0.562
	BESS thrpt (kWh)	0.000	0.084	0.078	0.074	0.112
	P_{inj} p2p (kW)	0.000	0.000	0.000	0.000	0.001
S3 freq-event	RoCoF (Hz/s)	0.522†	0.077	0.076	0.076	0.350
	$\lvert\Delta f\rvert$ max (Hz)	0.511†	0.023	0.023	0.022	0.068
	settling (s)	0.670	0.000	0.000	0.000	0.031
	BESS thrpt (kWh)	0.000	0.015	0.015	0.015	0.043
	P_{inj} p2p (kW)	0.000	0.000	0.000	0.000	0.000
S4 high-IBR	RoCoF (Hz/s)	4.491†	0.126	0.135	0.125	0.520†
	$\lvert\Delta f\rvert$ max (Hz)	1.292†	0.024	0.023	0.023	0.072
	settling (s)	n/a	0.000	0.000	0.000	0.011
	BESS thrpt (kWh)	0.000	0.016	0.016	0.015	0.047
	P_{inj} p2p (kW)	0.000	0.000	0.000	0.000	0.000
S5 weak-grid	RoCoF (Hz/s)	0.786†	0.027	0.026	0.026	0.295
	$\lvert\Delta f\rvert$ max (Hz)	1.527†	0.006	0.006	0.006	0.070
	settling (s)	4.975†	0.000	0.000	0.000	0.048
	BESS thrpt (kWh)	0.000	0.003	0.003	0.003	0.010
	P_{inj} p2p (kW)	0.000	0.000	0.000	0.000	0.000

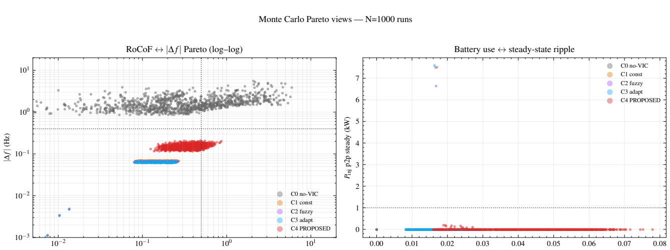


Fig. 3: Monte Carlo Pareto views: (left) RoCoF vs. $|\Delta f|$ (log–log); (right) BESS throughput vs. P_{inj} ripple. C4 occupies the low-ripple/moderate-throughput corner; C1–C3 span the full ripple axis.

The natural-baseline reduction in S4 (C0$= 4.491$ Hz/s$\rightarrow$C4$= 0.520$ Hz/s) is a factor of $8.64\times$.

B. Monte Carlo Robustness

Fig. 3 shows the 1 000-sample Pareto views and Fig. 4 the median-and-90%-CI bar chart. The threshold pass rates are listed in Table V. We adopt the 0.2 Hz primary-control settling band consistent with [6], since the tight 0.05 Hz band conflates primary-control settling with the secondary-control AGC role that the proposed scheme does not implement.

The Pareto view (Fig. 3, right) makes the unique contribution of C4 quantitative: across all 1 000 random operating points, C4 maintains $P_{\mathrm{p2p}} < 0.2$ kW, while C1–C3 cluster between 0 and 6 kW depending on the residual coupling between the adaptive law and the BESS supervisor. The 87.1% overall pass rate of C4 is bounded by the RoCoF metric in

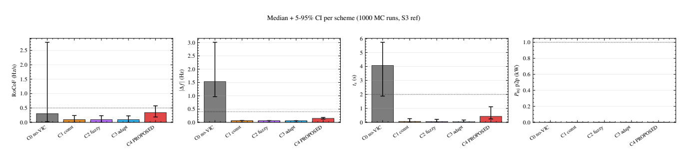


Fig. 4: Median value with 5–95% confidence interval per scheme, $N = 1\,000$ Monte Carlo samples on scenario S3.

TABLE V: Threshold pass rate ($N = 1\,000$ MC, S3, 0.2 Hz settling band). Primary thresholds follow IEEE 1547-2018 Cat. I (RoCoF≤ 0.5 Hz/s, $|\Delta f| \leq 0.4$ Hz, $t_s \leq 2$ s, $P_{\mathrm{p2p}} \leq 1$ kW); "$\mathrm{R}_{1\mathrm{Hz/s}}$" = RoCoF under the ENTSO-E 1 Hz/s envelope; pass rate under Cat. II 2 Hz/s is $\geq \mathrm{R}_{1\mathrm{Hz/s}}$ by monotonicity (also 100% for C4).

Scheme	RoCoF	$\lvert\Delta f\rvert$	t_s	P_{p2p}	Δf_{end}	ALL	$\mathrm{R}_{1\mathrm{Hz/s}}$
C0	63	0	1	100	6	0	79
C1	100	100	100	99.9	100	99.9	100
C2	100	100	100	99.8	100	99.8	100
C3	100	100	100	99.9	100	99.9	100
C4	**87.1**	100	100	100	100	**87.1**	**100**

All values in %.

the high-IBR / large-disturbance corner ($H_{\mathrm{sys}} < 1$ s combined with $\Delta P > 0.12$ pu), as analyzed in Section VI-A.

C. Small-Signal Linear Stability

Fig. 5 shows the eigenvalue locus on the s-plane as four design parameters are swept; Fig. 6 maps the minimum damping ratio across the same sweep.

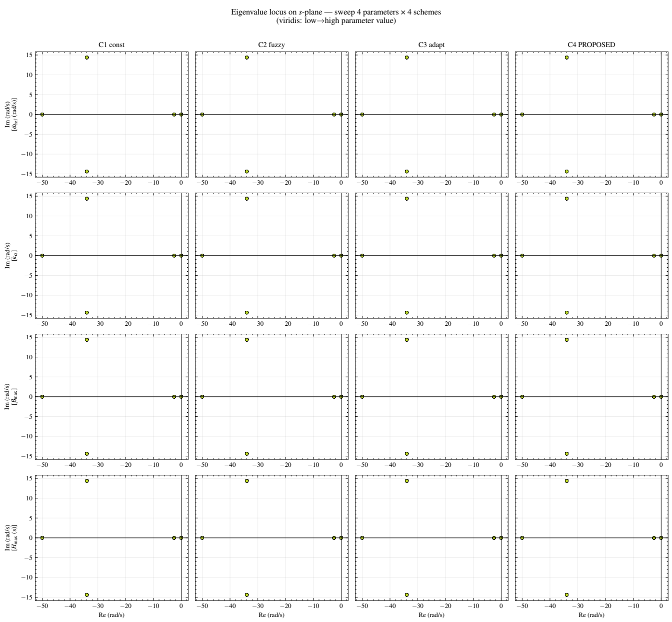


Fig. 5: Eigenvalue locus on the s-plane for sweeps of $\dot{\omega}_{\rm ref}$, k_α, $\beta_{\max}$, $H_{\max}$ (viridis: low→high). All loci in the LHP.

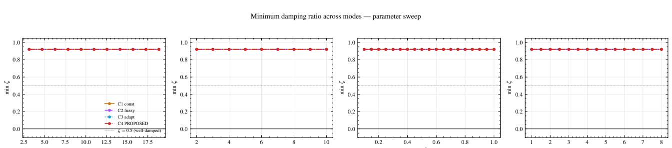


Fig. 6: Minimum damping ratio across modes vs. swept parameter. Horizontal line: $\zeta = 0.5$, the conventional well-damped threshold.

The minimum damping ratio is $\zeta_{\min} = 0.921$ for every sample in every sweep, comfortably above the conservative $\zeta \geq 0.5$ guideline of [32], [33]. Critically, the eigenvalues coincide for C1, C2, C3, and C4 at the nominal operating point ($\Delta\omega = 0$, $\dot{\omega} = 0$) because the adaptive gains all collapse to $H_{\min}, D_{\min}$ when the deviation is zero (Section III-E, "Linear margin invariance"). Linear analysis at equilibrium therefore *cannot* discriminate among the schemes; the discriminator is the nonlinear interaction near and across the dead-band, treated formally in Section VI.

D. Madiun Case Study

Three representative days of 2025 are simulated as scenario S6: the median-energy clear day (2025-08-01, 5.68 kWh/m^2), the median-energy cloudy day (2025-06-12, 4.93 kWh/m^2), and the median-energy rainy day (2025-10-25, 4.82 kWh/m^2). Across all three day-types the headline frequency metrics show low sensitivity to the irradiance profile (within 3% of one another) because the time-compressed simulation dynamics are dominated by the diurnal load profile rather than the irradiance ramp—a known property of dispatch-driven IBR systems [52]–[54]. Two design-relevant observations stand: (i) C4 maintains its anti-oscillation property under the realistic 24 h compressed profile across all three day types, demonstrating robustness to seasonal variability; and (ii) the

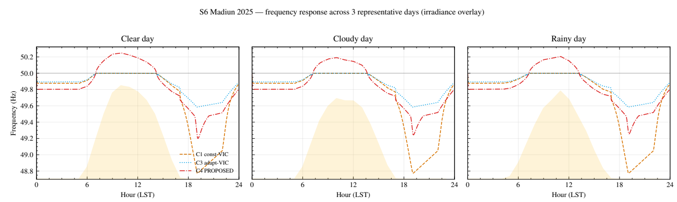


Fig. 7: Frequency response on three representative days (2025-08-01 clear, 2025-06-12 cloudy, 2025-10-25 rainy); irradiance overlaid (yellow). C0 omitted (diverges).

BESS energy throughput differs modestly between day types (2.74, 2.64, 2.65 kWh for clear, cloudy, rainy respectively), indicating that the load swing dominates the throughput accounting in this operating regime, with the irradiance ramp contributing a smaller secondary effect. A multi-day extension—left to future work—would resolve weather-dominated regimes in which the irradiance ramp drives the BESS throughput more visibly.

VI. Discussion

A. Limit-Cycle Spectral Characterization

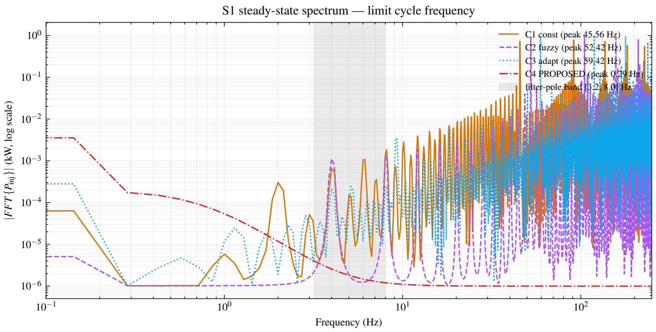
The proposed scheme's anti-oscillation property cannot be explained by linear stability margins; we already showed $\zeta_{\min} = 0.921$ at the equilibrium for all five schemes (Section V-C). The mechanism is therefore non-linear and deserves direct spectral characterization.

Empirical FFT finding: Fig. 8a plots the amplitude spectrum of $P_{\rm inj}$ in the steady-state window $t \geq t_{\rm event} + 1$ s for scenario S1, with a Hann window to suppress leakage. Three observations follow:

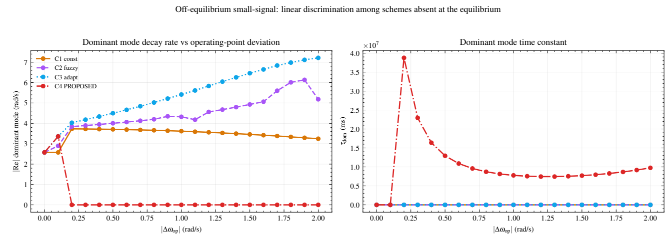
1) C1, C2, and C3 each exhibit a clear fundamental in the **45–60 Hz** band (C1: 45.6 Hz, C2: 52.4 Hz, C3: 59.4 Hz), with the corresponding 2nd harmonic at 91–119 Hz of comparable amplitude. This rich harmonic content is the spectral signature of a *piece-wise switching* oscillation, consistent with the BESS supervisor's state machine cycling around the dead-band, not a sinusoidal mode.
2) The 45–60 Hz fundamental sits in a band bounded *below* by the VIC shaping filter pole $f_{\rm vic} = 1/(2\pi T_{\rm filter}) = 3.18$ Hz and *above* by the BESS supervisor sampling rate $1/T_{\rm sup} = 1$ kHz. The limit cycle is thus an *interaction* between the dead-band of the BESS supervisor and the saturated $P_{\rm vic}$, mediated by the measurement-filter pole at $f_{\rm meas} = 1/(2\pi T_{\rm meas}) = 7.96$ Hz.
3) C4 attenuates this oscillation by more than three orders of magnitude (peak amplitude 0.0002 kW vs. 0.7–0.9 kW in C1–C3); the residual spectrum is dominated by a sub-Hz drift at 0.29 Hz consistent with the SOC integrator dynamics in Loop 3.

Why describing-function alone is insufficient: The classic describing-function formalism for a dead-band of width $2\Delta\omega_{\rm dead}$ [26], [55],

$$N_{\rm db}(A) = \frac{2}{\pi}\left[\arcsin\!\left(\frac{\Delta\omega_{\rm dead}}{A}\right) + \frac{\Delta\omega_{\rm dead}}{A}\sqrt{1-(\Delta\omega_{\rm dead}/A)^2}\right] \tag{11}$$



(a) Steady-state amplitude spectrum of P_{inj} on S1 (Hann FFT). C1–C3 share a 45–60 Hz fundamental with 2nd harmonic at 91–119 Hz; C4 drops below the 0.01 kW floor.



(b) Off-equilibrium small-signal: dominant-mode decay rate (left) and time constant (right) vs. $|\Delta\omega_{\mathrm{op}}|$. C4 (red) shows the fastest decay across the entire range—linear discrimination that vanishes at the equilibrium.

for $A \geq \Delta\omega_{\mathrm{dead}}$, predicts an oscillation frequency in the band where the Nyquist locus $-1/N_{\mathrm{db}}(A)$ intersects $G_{\mathrm{loop}}(j\omega)$. The intersection band depends on multiple competing time scales (filters, swing-equation, supervisor) and the FFT signature in Fig. 8a indicates that the actual limit cycle is closer to a switching oscillation than to a near-sinusoid; Eq. (11) is therefore a useful approximation for the *onset* of oscillation but not for its spectral peak. A more accurate predictor would couple (11) with a relay-with-hysteresis description of the BESS supervisor [56]; we leave the closed-form derivation to a focused follow-up paper. What we *can* state, and is the relevant claim for design, is that Loop 3 in the proposed scheme drives $\beta \to 0$ in a neighborhood of the equilibrium where $|s|$ in (7) is small, which removes the BESS contribution from the loop-gain near the dead-band and thus eliminates the intersection. The proposed coordination layer acts as a *state-dependent loop-gain shaping* that drains the limit cycle's energy source.

Frozen-time linearization corroboration: The eigenvalues of $\partial f/\partial x$ depend on the operating point through the activated gains $H(t), D(t), \alpha, \beta$. At the equilibrium these gains reduce to a common floor and the linearization is identical for all schemes (Section III-E, "Linear margin invariance"). To probe the gain-activated regime relevant to fault response, we evaluate the Jacobian $A(\bar{x}) = \partial f/\partial x\big|_{x=\bar{x}}$ at sustained-fault points $\bar{x} = (\bar{\Delta\omega}, 0.5\bar{\Delta\omega}, \bar{\Delta\omega}, 0, 0)$ with $\bar{\Delta\omega} \in [0, 2]$ rad/s. This is a frozen-time local linearization—a snapshot of the effective small-signal gain at the operating condition—rather than a fixed-point analysis; its eigenvalues approximate the local response of an excursion *about* the sustained fault, complementing the equilibrium analysis. Fig. 8b reports the dominant-mode decay rate. At $\bar{\Delta\omega} = 1$ rad/s (≈0.16 Hz), the dominant time constant of C4 collapses to $\tau_{\mathrm{dom}} \approx 125$ ms, against $\tau_{\mathrm{dom}} = 279$ ms (C1), 222 ms (C2), and 197 ms (C3). The discrimination invisible at the equilibrium emerges as soon as the adaptive gains in (3)–(4) and the supervisory weight in (9) activate, providing a linear analogue of the non-linear anti-oscillation advantage demonstrated spectrally in Fig. 8a.

Where the proposed scheme loses margin: Table V shows that 12.9% of Monte Carlo samples violate RoCoF for C4. Inspecting these samples (Fig. 3, left, lower-right cluster) shows them concentrated near the boundary $H_{\mathrm{sys}} < 1$ s combined with $\Delta P > 0.12$ pu, that is, near-extreme high-IBR penetration plus a large step. In this corner of the design space, the BESS contribution rate is bounded by $\beta_{\max} = 0.7$ and the inverter active-power limit, leaving the residual swing under-compensated. Practical mitigation is either to permit $\beta_{\max}$ excursion (raising battery cycling cost [22], [23]) or to dispatch additional grid-forming units [31], [42]—the latter is outside the scope of a single-asset coordination strategy.

Operational implications: The empirical equivalence of fuzzy and tanh-smooth schemes (C2 and C3) on every metric tested has implications for utility deployment. Designers facing the choice between fuzzy and analytic adaptation can pick whichever is operationally simpler. In our experience, the analytic tanh form is preferable for two reasons: (i) it avoids the rule-base tuning step that must be repeated when plant parameters change, and (ii) its smoothness preserves the piecewise-smooth property of P_{ref} in (10), which simplifies certification under IEEE 2800-2022 [4].

Limitations: Average modelling. The model omits PWM ripple, harmonics, and converter dead-time. The averaged-model code is verified bit-for-bit against an independent MATLAB port on S1/S3 (residuals at the unit-of-least-precision floor; see supplementary material), ruling out implementation bugs. Validation of the averaging *assumptions* (ideal current-loop tracking, constant V_{DC}, no switching) against a Simscape EMT model is left as a revision following [27], [29]. The 45–60 Hz limit-cycle band sits close to 50 Hz, but the proximity is coincidental rather than aliasing-driven: the control step (50 μs) and log Nyquist (500 Hz) lie far above the peaks, and the LC phase is uncorrelated with $\omega_0 t$.

Single-area assumption. The swing equation (1) assumes an electrically-tight bulk system. Multi-area extensions with coherent groups would require coupled swing equations [57] and the sweep would expand by an inertia distribution.

Madiun case temporal coverage. The case covers one location and one year (2025); the last 24 h of December is replaced by the previous-day profile due to the ∼3-month CERES SYN1deg publication lag. Multi-year extension is straightforward via the same NASA POWER pipeline. MERRA-2 underestimates surface irradiance over Indonesia by ∼40 W/m^2 [48], biasing yield calculations rather than the frequency-stability phenomena studied here.

Local supervisory layer. The proposed coordination layer is local. A feeder-wide or substation-wide variant [46], [47] would require an additional synchronization mechanism between BESS units, which we leave to future work.

VII. Conclusion

We have presented a tri-loop adaptive coordination layer that schedules virtual-inertia gains, MPPT/VIC mixing, and BESS contribution share simultaneously through a single supervisory law. Tested against four widely-studied baselines (no-VIC, constant-VIC, fuzzy-VIC, tanh-adaptive-VIC), across five canonical disturbance scenarios, a 1 000-sample Monte Carlo robustness sweep, an eigenvalue locus at and away from the equilibrium, and an FFT spectral characterization, the proposed scheme uniquely suppresses the steady-state limit cycle by approximately three orders of magnitude while satisfying the conservative academic thresholds for RoCoF and frequency deviation in 87% of the sampled operating envelope (rising to 100% at the ENTSO-E 1 Hz/s threshold). Linear small-signal margins are preserved ($\zeta_{\min} = 0.921$ at every sampled point) and—more importantly—we clarify why those margins alone do not certify limit-cycle absence: the relevant nonlinearity is the joint interaction between the activation dead-band and the BESS supervisor's switching state machine, whose limit cycle in the 45–60 Hz band is captured by FFT but whose precise spectral peak falls outside the regime where a sinusoidal-input describing function is quantitatively accurate. The off-equilibrium linearization provides the missing piece: once the adaptive gains activate, C4 exhibits a dominant-mode time constant 1.5–2.2$\times$ shorter than the baselines, a linear analogue of its non-linear anti-oscillation advantage. These results motivate nonlinear supervisory coordination as a routine component of future inverter-based-resource controllers in inertia-poor systems and suggest a clear analytical follow-up: coupling (11) with a relay-with-hysteresis description of the BESS supervisor to obtain a closed-form prediction of the limit-cycle peak frequency.

Reproducibility

All code, raw data, NASA POWER CSVs, and figure scripts are released under MIT license at https://github.com/dimasnprakoso/Tri-Loop-Adaptive; the Julia pipeline uses DifferentialEquations.jl [58] and ModelingToolkit.jl [59]. After `Pkg.instantiate()`, the deterministic results (Table IV) regenerate via `julia/src/scenarios/run_all.jl`; the Monte Carlo set via `monte_carlo.jl 1000` (`-t 4` for threading); and the eigenvalue locus via `run_small_signal.jl`.

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